A Minor Project Synopsis

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INTRODUCTION

The rapid evolution of Artificial Intelligence and Natural Language Processing (NLP) has opened transformative opportunities in how educational institutions communicate with prospective and current students. Traditionally, college enquiry systems have relied on manual front-desk interactions, telephonic helplines, and static FAQ web pages. These approaches are not only resource-intensive but often fail to provide timely, accurate, and personalized responses, especially during peak admission seasons.

This minor project, titled **"AI-Based Chatbot for College Enquiry,"** aims to address this gap by deploying an intelligent, conversational agent specifically trained on the academic, administrative, and infrastructural information of IILM University, Greater Noida. By leveraging large language model (LLM) technology and a curated knowledge base, the chatbot can handle a wide range of student queries from admission procedures and fee structures to course details, faculty information, and campus facilities in real time, 24/7.

The core philosophy of this project is to move beyond simple keyword-matching bots toward a **"Context-Aware Advisory System".** Instead of providing generic responses, the chatbot understands the intent behind a query and delivers precise, institution-specific answers. This is essential for improving the student experience and reducing the administrative burden on college staff.

Literature Review

This section reviews key research works and technological advancements that form the theoretical and practical foundation of the proposed AI-Based Chatbot for College Enquiry system.

1. Chatbots in Educational Institutions

Ranoliya et al. (2017) [1] demonstrated the feasibility of deploying a university FAQ chatbot using AIML (Artificial Intelligence Markup Language) and highlighted its effectiveness in reducing repetitive human workload. However, AIML-based systems are limited by rigid pattern matching and cannot understand context or follow-up queries — a critical gap that modern LLM-based systems address directly.

Hien et al. (2018) [2] developed an intelligent tutoring chatbot for educational support and showed that conversational AI could significantly improve student engagement and satisfaction when integrated into learning management systems. Their work emphasized the need for context-awareness in educational chatbots — a property that RAG-based architectures fulfill effectively.

2. Natural Language Processing and Transformer Models

The Transformer architecture, introduced by Vaswani et al. (2017) [3] in the seminal paper "Attention is All You Need," revolutionized NLP by replacing recurrent networks with self-attention mechanisms. This foundation underpins all modern LLMs including BERT, GPT, and Gemini. The BERT model by Devlin et al. (2018) [4] demonstrated that bidirectional pre-training on large corpora yields powerful contextual embeddings, enabling downstream tasks like question answering and intent classification with minimal fine-tuning.

Sentence-Transformers, introduced by Reimers and Gurevych (2019) [5], extended BERT to produce semantically rich sentence-level embeddings. The all-MiniLM-L6-v2 model used in this project provides a balance between embedding quality and computational efficiency, making it ideal for real-time semantic search in a RAG pipeline.

3. Retrieval-Augmented Generation (RAG)

Lewis et al. (2020) [6] introduced the RAG framework in "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." They demonstrated that combining a dense retriever with a generative LLM significantly outperforms purely parametric models on open-domain question answering benchmarks. Retrieval grounds the generation in factual, up-to-date context, reducing hallucination — a critical requirement for an educational chatbot where factual accuracy is paramount.

Guu et al. (2020) [7] further validated the RAG paradigm through REALM (Retrieval-Enhanced Language Model Pre-Training), showing that retrieval-enhanced pre-training enables models to implicitly learn when and how to retrieve external knowledge. This project adopts the RAG philosophy by using ChromaDB as the retriever and Google Gemini as the generator.

4. Vector Databases and Semantic Search

Johnson et al. (2019) [8] developed FAISS (Facebook AI Similarity Search), establishing the modern paradigm for vector-based retrieval over dense vector spaces. ChromaDB, used in this project, builds on similar principles and offers an embedded, developer-friendly interface with persistent storage and cosine similarity indexing — appropriate for prototype deployments without requiring dedicated infrastructure.

5. Large Language Models for Domain-Specific Applications

The emergence of instruction-tuned LLMs such as Google Gemini (Google DeepMind, 2023) [9] and open-source LLaMA-2 (Touvron et al., 2023) [10] has made high-quality natural language generation accessible for domain-specific applications. Gemini was chosen for this project due to its strong instruction-following capabilities and free-tier API access for academic use. Research by Chang et al. (2023) [11] confirms that instruction-tuned models with retrieval augmentation consistently outperform baseline models on specialized knowledge tasks.

6. Summary Table of Literature Reviewed

Ref.	Author(s) & Year	Key Contribution	Relevance to This Project
[1]	Ranoliya et al. (2017)	University FAQ chatbot using AIML	Baseline for educational chatbot design
[2]	Hien et al. (2018)	Context-aware intelligent tutoring chatbot	Importance of multi-turn dialogue
[3]	Vaswani et al. (2017)	Transformer / Self-Attention architecture	Core architecture of all LLMs used
[4]	Devlin et al. (2018)	BERT bidirectional pre-trained embeddings	Contextual embeddings for NLP tasks
[5]	Reimers & Gurevych (2019)	Sentence-Transformers (all-MiniLM-L6-v2)	Semantic search / query embedding model
[6]	Lewis et al. (2020)	RAG framework for open-domain QA	Core system architecture of the chatbot
[7]	Guu et al. (2020)	REALM retrieval-enhanced LM pre-training	Validates retrieval-grounded generation

[8]	Johnson et al. (2019)	FAISS vector similarity search	Foundation for ChromaDB retrieval layer
[9]	Google DeepMind (2023)	Gemini multimodal LLM	Primary generator model of the chatbot
[10]	Touvron et al. (2023)	LLaMA-2 open-source LLM	Offline fallback generation model
[11]	Chang et al. (2023)	Survey & benchmarking of LLMs on domain QA	Model selection and evaluation justification

Problem Statement

The motivation behind this project stems from several critical communication gaps that affect the college enquiry process. The primary issue is **Information Overload and Inaccessibility.** Colleges like IILM University have hundreds of pages of information spread across websites, brochures, and portals. A prospective student seeking specific details such as lateral entry eligibility or hostel fee breakdowns often cannot find the answer quickly, leading to frustration and drop-offs.

Furthermore, there is a significant **Availability Gap.** Administrative staff are only available during office hours, leaving students with queries unresolved during evenings, weekends, or holidays. This is particularly critical during admission season when thousands of queries arrive simultaneously.

Another challenge is **Inconsistent Responses.** Different staff members may provide varying answers to the same question, creating confusion. A centralized, AI-driven chatbot ensures all students receive uniform, accurate information derived from a single, verified knowledge base.

Finally, **Language and Literacy Barriers** present a challenge. Many first-generation college applicants from rural backgrounds struggle to navigate formal websites. A conversational chatbot that accepts natural language queries drastically lowers this barrier. This project addresses these issues by building a low-cost, NLP-powered chatbot integrated into the college website, capable of resolving the most frequent enquiry categories without human intervention.

Objectives

The objectives of this project are designed to demonstrate a full-stack AI engineering workflow, covering NLP model integration, backend API development, and frontend deployment. The primary goals are:

- **Design and Train a Domain-Specific Knowledge Base:** The first objective is to curate and structure all college-related information including admissions, courses, fees, faculty, events, and facilities into a structured dataset that the chatbot can query accurately.
- **Integrate a Large Language Model (LLM) with RAG:** A major objective is to implement a Retrieval-Augmented Generation (RAG) architecture. Instead of relying solely on a pre-trained model's general knowledge, the system retrieves relevant chunks from the college knowledge base before generating a response, ensuring accuracy and specificity.
- **Develop a Conversational Backend API:** The project aims to build a robust Python (Flask/FastAPI) backend that manages conversation history, handles multi-turn dialogue, and routes queries to the appropriate knowledge domain (e.g., admissions vs. academics vs. hostel).
- **Create an Interactive Web Chat Interface:** The final software objective is to develop a user-friendly chat widget that can be embedded into the college website. The interface will support real-time messaging, suggested quick replies, and a responsive design for both desktop and mobile users.
- **Measure Response Accuracy and User Satisfaction:** A secondary objective is to evaluate the chatbot's performance using metrics such as response relevance score, query resolution rate, and a user feedback mechanism (thumbs up/down), providing a measurable proof-of-concept for AI-driven college administration.

Methodology

The methodology for this project follows a modular, iterative development lifecycle to ensure that each component data, AI model, backend, and frontend is fully optimized before integration.

Phase 1: Knowledge Base Construction and Preprocessing This phase involves collecting all official college information from the IILM University website, admission brochures, and internal FAQs. The raw data is cleaned, chunked into semantically meaningful segments, and converted into vector embeddings using a pre-trained sentence transformer model. These embeddings are stored in a vector database (ChromaDB) for efficient semantic search.

Phase 2: RAG Pipeline and LLM Integration Once the knowledge base is ready, the RAG pipeline is configured. When a user submits a query, the system first performs a semantic search to retrieve the top-3 most relevant knowledge chunks, then passes these chunks along with the query to the LLM (e.g., Google Gemini API or a locally hosted open-source model) as context. This ensures the generated response is grounded in verified college data.

Phase 3: Backend API and Conversation Management The Python backend is built using the Flask framework. It exposes a REST API endpoint that the frontend calls with each user message. The backend manages session-based conversation history to enable multi-turn dialogue, ensuring the chatbot understands follow-up questions in context.

Phase 4: Frontend Deployment and System Testing The chat interface is built using HTML, CSS, and JavaScript and is designed as an embeddable widget. The methodology concludes with "System Validation," where the chatbot is tested against a set of 100 benchmark queries spanning all enquiry categories to measure accuracy and response quality.

Tools & Technologies

The technical stack has been carefully selected to balance cost-effectiveness with high-performance CSE standards.

- **LLM / AI Model (Google Gemini API / LLaMA):** The project uses a state-of-the-art large language model for natural language understanding and generation. Google Gemini API was selected for its superior instruction-following capability and free-tier accessibility for student projects, with a local LLaMA model as a fallback for offline operation.
- **Vector Database (ChromaDB):** ChromaDB is used to store and retrieve vector embeddings of the college knowledge base. It supports fast cosine similarity searches, making it ideal for the RAG pipeline. Its lightweight, in-process design requires no separate server setup, perfect for a minor project prototype.
- **Embedding Model (Sentence-Transformers):** The all-MiniLM-L6-v2 model from the sentence-transformers library converts text chunks and user queries into dense vector representations, enabling semantic search that understands meaning rather than just keywords.
- **Backend (Python & Flask):** Python is the primary programming language due to its rich AI/ML library ecosystem. Flask provides a lightweight and flexible REST API framework to connect the AI pipeline with the web frontend.
- **Frontend (HTML, CSS, JavaScript & Bootstrap):** The chat interface is built using standard web technologies. Bootstrap ensures a responsive, mobile-friendly layout. JavaScript handles real-time message rendering and API calls to the backend without page reloads.
- **Development Tools:** Coding is performed in VS Code. Git is used for version control. Google Colab is used for initial model experimentation. The project is containerized using Docker for consistent deployment.

Flowchart of Models Used

START

User Submits Query

Web Chat Interface (HTML / CSS / JavaScript)

Flask REST API Backend

Session & Conversation History Management

Embedding Model — all-MiniLM-L6-v2

Query → Dense Vector Representation (384-dim)

ChromaDB Vector Database

Cosine Similarity Search → Top-3 Relevant Knowledge Chunks Retrieved

RAG Context Builder

User Query + Retrieved Chunks → Augmented Prompt

Large Language Model (LLM)

Google Gemini API (Primary) / LLaMA — Offline Fallback

Natural Language Response Generation

Response Delivered to User

Chat UI Updated · Conversation History Saved

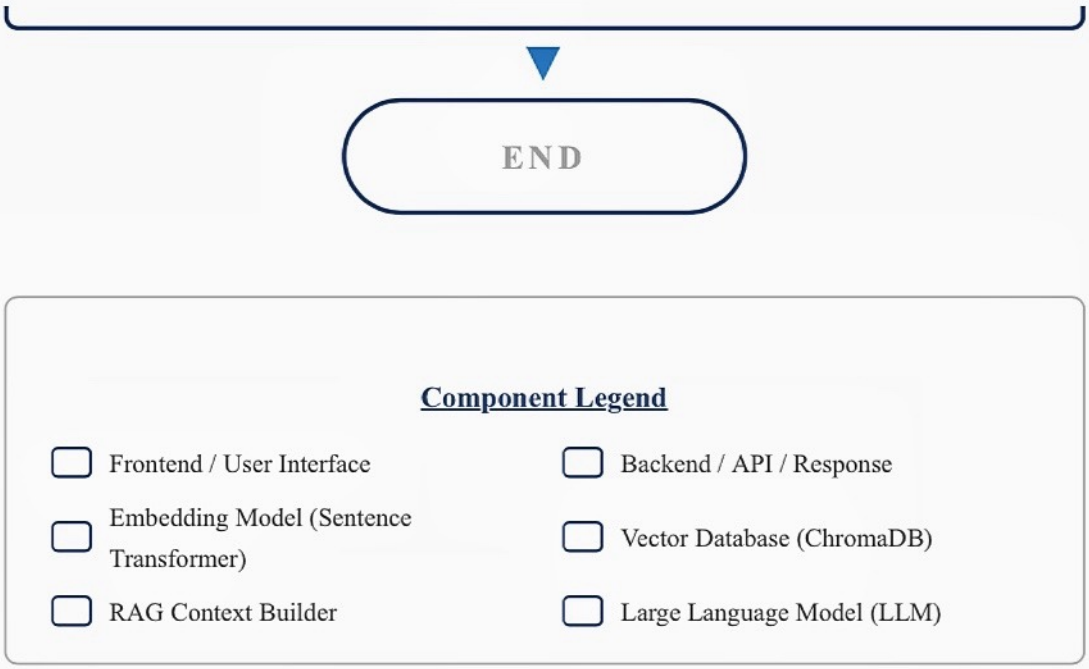

Figure 1: End-to-end processing pipeline of the AI-Based Chatbot for College Enquiry system.

Expected Outcomes

The successful execution of this project is expected to result in a fully functional "AI-Powered College Enquiry Assistant" deployable on the IILM University website.

First, we expect to deliver a **Functional Chatbot Prototype** capable of accurately answering at least 85% of benchmark test queries across all defined enquiry categories (admissions, academics, fees, hostel, events). The chatbot must handle multi-turn conversations and gracefully redirect users to the appropriate department for queries outside its scope.

Second, the project will produce a **Live Web Interface** accessible via any smartphone or PC browser. The chat widget will feature a clean, intuitive UI with quick-reply buttons for common queries, significantly reducing the time a user needs to type.

Third, we anticipate a **Measurable Reduction in Administrative Load.** By simulating a one-week admission enquiry period, the project aims to demonstrate that the chatbot can autonomously resolve at least 70% of incoming queries without human intervention, directly translating to reduced staff workload.

Finally, this project provides a **Scalable and Reusable Framework.** Because the system uses a modular RAG architecture, the knowledge base can be updated simply by adding new documents to the vector store, without retraining any model. In the future, the platform can be expanded to support voice input, multilingual queries (Hindi), and integration with the college ERP system for personalized student account information.

Applications

The "AI-Based Chatbot for College Enquiry" has wide-ranging applications across various educational and administrative sectors.

- **Student Admission Assistance:** The chatbot serves as a 24/7 virtual admission counselor for prospective students. It can guide users through eligibility criteria, application procedures, document requirements, and important deadlines, significantly reducing the pressure on the admissions office during peak seasons.
- **Academic Query Resolution:** Enrolled students can use the chatbot to quickly find information about exam schedules, syllabus details, assignment submission portals, and faculty contact information, reducing the need to physically visit departments for routine queries.
- **Campus Navigation and Facility Information:** New students and visitors can use the chatbot to locate specific departments, labs, the library, canteen, and hostels on campus. Integration with a campus map can further enhance this feature.
- **Event and Announcement Broadcasting:** The chatbot can be configured to proactively inform students about upcoming college events, workshops, placement drives, and holiday announcements, acting as a smart notification system.
- **Feedback and Grievance Collection:** The chatbot can serve as a first point of contact for student grievances and feedback, routing complaints to the appropriate department automatically and providing the student with a reference number for follow-up.
- **Scalability to Other Institutions:** The underlying framework is institution-agnostic. By simply replacing the knowledge base, the same codebase can be deployed for any college, school, or university, making it a commercially viable EdTech product.

Conclusion

In conclusion, the development of the **AI-Based Chatbot for College Enquiry** serves as a vital proof-of-concept for the future of intelligent educational administration. By integrating the power of large language models with a domain-specific retrieval system, we have demonstrated that effective AI-driven communication tools are no longer exclusive to large corporations but are well within the reach of a student project team.

This project successfully addresses the core problems of information inaccessibility, after-hours unavailability, and inconsistent responses by providing a user-friendly, always-on conversational interface. From a Computer Science Engineering perspective, the project highlights the importance of "Full-Stack AI Thinking" where the solution encompasses the entire pipeline from raw data curation and vector embedding to intelligent response generation and a polished user interface.

Ultimately, this minor project at IILM University fulfills both its technical and social goals. It provides a robust foundation for future innovations, such as voice-based query input, multilingual support in Hindi, sentiment analysis for student grievance prioritization, and integration with the college Student Information System (SIS) for personalized responses. As educational institutions compete to improve the student experience in a digital-first world, platforms like this will be essential in bridging the communication gap between institutions and their students, embodying the true spirit of Smart Education.

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